Accelerating Polynomial System Solving via Factorization and Unsatisfiable-Core Learning

Ilay Menahem[a], Michael Shalyt[a] and Ido Kaminer[a,b]

[a]Andrew and Erna Viterbi Faculty of Electrical and Computer Engineering, Technion – Israel Institute of Technology, Haifa, 3200003, Israel
[b]Department of Materials Science and Engineering, Technion – Israel Institute of Technology, Haifa, 3200003, Israel

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ABSTRACT

Polynomial systems are a fundamental language for describing algebraic and geometric problems in mathematics and its applications. Beyond finding individual solutions, one often seeks global properties of the solution space, such as its dimension and irreducible components, or structural properties and decompositions of the underlying polynomial ideal. For large polynomial systems with many variables and high degrees, determining these global properties can become intractable. Leading approaches include variants of Buchberger's algorithm with syzygy, resultants, and cylindrical algebraic decomposition. However, even the most modern approaches still often fail on practical problems or slight perturbations of standard benchmarks. Here we introduce a framework that divides a polynomial system into simpler subproblems by interreducing and factoring its equations. The framework only assumes that the solver used on each subproblem describes satisfiable (SAT) branches and certifies unsatisfiable (UNSAT) ones. If the solver also returns an UNSAT core, the framework can use it to prune later branches, following the same idea as conflict-driven clause learning (CDCL). When individual branch computations exceed time limits, the framework refines stalled branches through recursive variable splitting, illustrated here by zeroing ($x = 0$ versus $x \neq 0$) to isolate simpler sub-varieties. In our experiments, we use Gröbner-basis solvers and apply the method to a system arising from research in quantum field theory for which no standard method in our comparison succeeds. Together, these ideas offer a path toward the exact analysis of previously inaccessible polynomial systems.

1. Introduction

Systems of multivariate polynomial equations appear whenever a symbolic model contains nonlinear algebraic constraints. Applications include robotics, coding theory, optimization, statistics, and physics (Sturmfels, 2002). In many of these problems, an isolated solution is not enough. One needs the dimension of the solution set, its irreducible components, or the exact relations that hold on every solution. These are questions about the variety and its defining ideal, rather than about a single point (Cox et al., 2015; Kreuzer and Robbiano, 2000).

Elimination by resultants provided an early systematic way to attack polynomial systems, but the introduction of Gröbner bases gave a common algorithmic framework for elimination, ideal membership, dimension, and related tasks (Cox et al., 2015; Kreuzer and Robbiano, 2000). Buchberger developed this framework in his 1965 thesis (Buchberger, 2006). Later algorithms changed how the expensive reductions are organized. F4 replaces individual polynomial reductions with sparse linear algebra, while F5 uses signatures to avoid many reductions to zero (Faugère, 1999, 2002). Primary-decomposition algorithms use Gröbner bases and factorization to separate components (Gianni et al., 1988). Numerical algebraic geometry takes a different route: homotopy continuation computes isolated roots numerically, while witness sets represent positive-dimensional components (Sommese et al., 2001; Sommese and Wampler, 2005).

Recent work has also explored machine-learning methods for polynomial systems. Peifer et al. trained a reinforcement-learning policy for pair selection in Buchberger's algorithm (Peifer et al., 2020). Kera et al. trained transformers to compute Gröbner bases directly for a restricted class of zero-dimensional radical ideals (Kera et al., 2024).

Despite this progress, large systems can remain inaccessible.

Here, we introduce a solver-agnostic framework that breaks down a polynomial system into smaller subproblems by interreducing and factoring its equations. The framework enumerates the minimal hitting sets of the resulting factor

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sets; each hitting set contains at least one factor from every equation and therefore defines one branch of the original solution set. An exact branch solver then either describes that branch, including any positive-dimensional families, or certifies that it is UNSAT. Since the worst-case complexity of computing a Gröbner basis is doubly exponential in the number of variables (Mayr and Meyer, 1982), breaking down a large system into smaller subproblems can be substantially faster than solving the original system directly.

To avoid repeated work, the framework learns certified UNSAT cores and prunes later hitting sets that contain them, adapting ideas from polynomial UNSAT-core extraction and conflict-driven learning (Marques-Silva et al., 2009; Marques-Silva and Sakallah, 1999; Sun et al., 2016).

If a factor branch remains too expensive for the chosen solver, the framework can refine it by splitting a selected variable into the cases $x = 0$ and $x \neq 0$. This heuristic split can expose additional algebraic structure for the branch solver without losing any solutions. While this is one of many possible splitting criteria, it proves particularly effective on the quantum field theory problem that we investigate.

We prove that refinement and variable zeroing terminate and preserve the full solution set. We compare the resulting implementation with direct Gröbner-basis computation, symbolic decomposition, and numerical homotopy on generated benchmarks. We also apply the framework to a system from quantum field theory: with the `facstd` backend, the framework completes the computation, whereas standard standalone solvers timed out.

2. Preliminaries and notation

2.1. Polynomial systems, ideals, and varieties

Let K be a field, let $\overline{K}$ be its algebraic closure, and write $R = K[x_1, \dots, x_n]$. The input is a finite system $P = \{p_1, \dots, p_m\} \subseteq R$ of nonzero polynomials. Unless stated otherwise, a solution is a point in $\overline{K}^n$. For a finite set $Q \subseteq R$, its solution set is

$$V(Q) = \{a \in \overline{K}^n : q(a) = 0 \text{ for every } q \in Q\}.$$

We call Q satisfiable (SAT) if $V(Q) \neq \emptyset$ and unsatisfiable (UNSAT) if $V(Q) = \emptyset$.

The ideal generated by Q, denoted $\langle Q \rangle$, consists of all sums $\sum_{q \in Q} h_q q$ with $h_q \in R$.

For $X \subseteq \overline{K}^n$, its vanishing ideal $I(X)$ contains every polynomial in R that vanishes on all of X. The radical of an ideal J is $\sqrt{J} = \{g \in R : g^r \in J \text{ for some } r \geq 1\}$. Hilbert's Nullstellensatz gives

$$I(V(Q)) = \sqrt{\langle Q \rangle}.$$

Thus $V(Q) = \emptyset$ exactly when $\langle Q \rangle = \langle 1 \rangle$. These facts connect solution sets with computations in polynomial ideals (Cox et al., 2015; Kreuzer and Robbiano, 2000).

2.2. Monomial orders and Gröbner bases

Let $\mathbb{M}_n = \{x^\alpha : \alpha \in \mathbb{N}^n\}$ be the monomials of R. A monomial order $\prec$ is a total order on $\mathbb{M}_n$ such that every nonempty subset has a least element and

$$u \prec v \quad \Longrightarrow \quad uw \prec vw \qquad (u, v, w \in \mathbb{M}_n).$$

For a nonzero polynomial p, the leading monomial $\mathrm{LM}(p)$ is the greatest monomial of p under $\prec$, $\mathrm{LC}(p)$ is its coefficient, and $\mathrm{LT}(p)$ is its leading term.

For an ideal J, define its leading-monomial ideal by

$$\mathrm{LM}(J) = \langle \mathrm{LM}(f) : f \in J \setminus \{0\} \rangle.$$

A finite set $G \subseteq J$ is a Gröbner basis of J if

$$\mathrm{LM}(J) = \langle \mathrm{LM}(g) : g \in G \rangle.$$

A finite set Q is *interreduced* if every element is monic and no term of one element is divisible by the leading monomial of another. Algorithm 1 interreduces a finite set of nonzero polynomials.

Algorithm 1 Interreduction

Require: A finite set $Q \subseteq R \setminus \{0\}$ and a monomial order $\prec$
Ensure: An interreduced set that generates $\langle Q \rangle$

```
while there exist distinct p, q ∈ Q and a term cu of p with LM(q) | u do
    r ← p − (cu / LT(q)) q
    Q ← Q \ {p}
    if r ≠ 0 then
        Q ← Q ∪ {r}
    end if
end while
Q ← {p / LC(p) : p ∈ Q}
return Q
```

Each loop iteration cancels one monomial u and introduces only monomials strictly smaller than u. The multiset extension of the monomial order is well-founded, so the loop terminates. Interreduction preserves $\langle Q \rangle$ and $V(Q)$.

Every ideal has a unique reduced Gröbner basis for a fixed monomial order. The reduced basis is $\{1\}$ if and only if $J = \langle 1 \rangle$. Therefore, a Gröbner basis computation can describe a branch and detect when it is UNSAT (Cox et al., 2015; Kreuzer and Robbiano, 2000).

An UNSAT core of a system Q is a subset $U \subseteq Q$ with $V(U) = \emptyset$. An UNSAT core is minimal if there is no $U' \subsetneq U$ such that $V(U') = \emptyset$. Any system containing U is also UNSAT, since adding equations cannot introduce solutions.

2.3. Factor sets and hitting sets

Fix a factorization of each input polynomial over K:

$$p_i = a_i \prod_{f \in C_i} f^{e_{i,f}},$$

where $a_i \in K \setminus \{0\}$, C_i contains the distinct monic, nonconstant irreducible factors of p_i, and $e_{i,f} \geq 1$. Multiplicity does not affect the zero set. A point satisfies $p_i = 0$ exactly when at least one factor in C_i vanishes at that point.

A hitting set for $C_1, \ldots, C_m$ is a set $H \subseteq \bigcup_i C_i$ such that $H \cap C_i \neq \emptyset$ for every i. It is minimal if no proper subset is also a hitting set. We write $\mathcal{H}(P)$ for the family of minimal hitting sets obtained from P. Each $H \in \mathcal{H}(P)$ defines a branch in which all factors in H vanish (Berge, 1989).

2.4. Encoding nonzero variables

Let $N \subseteq \{x_1, \ldots, x_n\}$ be the variables required to be nonzero. Here are two ways to encode these constraints as polynomial equations.

One encoding introduces a separate auxiliary variable s_x for each $x \in N$ and works in the extended ring $R_{\text{sep}} = R[s_x : x \in N]$. For each $x \in N$, define the polynomial

$$\sigma_x = 1 - s_x x, \tag{1}$$

and impose the equation $\sigma_x = 0$.

The second encoding uses one auxiliary variable s, works in $R_{\text{prod}} = R[s]$, and combines all nonzero constraints into the polynomial

$$\sigma_N = 1 - s \prod_{x \in N} x, \tag{2}$$

with the equation $\sigma_N = 0$.

Let $Q \subseteq R$ and let π denote the appropriate coordinate projection onto the original $x_1, \ldots, x_n$ coordinates. In the first encoding,

$$\pi\left(V\left(Q \cup \{\sigma_x : x \in N\}\right)\right) = \{a \in V(Q) : a_x \neq 0 \text{ for every } x \in N\}.$$

Indeed, $\sigma_x = 0$ has a solution s_x exactly when $a_x \neq 0$. Similarly, the product encoding satisfies

$$\pi\left(V\left(Q \cup \{\sigma_N\}\right)\right) = \{a \in V(Q) : a_x \neq 0 \text{ for every } x \in N\}.$$

3. Factor branching framework

3.1. Framework description

Polynomial-ideal computations can be intractable even for small systems (Mayr and Meyer, 1982). Thus, decomposing a system into smaller and simpler subproblems may make the individual solver calls easier.

We break a system into smaller subproblems by alternating two operations: interreducing the current subproblem, and factoring each generator to enumerate the minimal hitting sets of the factor sets. Then again do the same on each hitting set, until the subproblems can't be further broken down.

Here is the pseudocode for this refinement of a polynomial system Q into a smaller set of subproblems.

Algorithm 2 Refinement

Require: A finite polynomial system Q and a collection A of normalized systems visited on the current recursion path

$Q \leftarrow Q \setminus \{0\}$
$\widetilde{Q} \leftarrow \textsc{Interreduce}(Q)$ ▷ Optional; otherwise $\widetilde{Q} \leftarrow Q$
Factor each generator of $\widetilde{Q}$
$\mathcal{H} \leftarrow$ the minimal hitting sets of the factor sets
$T \leftarrow \emptyset$
for all $H \in \mathcal{H}$ **do**
    if $H = \widetilde{Q}$ or $H \in A$ **then**
        $T \leftarrow T \cup \{H\}$
    else
        $T \leftarrow T \cup \textsc{Refine}(H, A \cup \{\widetilde{Q}\})$
    end if
end for
return T

Let $\mathcal{L}(P)$ denote the terminal hitting sets obtained from P. For each $H \in \mathcal{L}(P)$, the branch solver must either describe $V(H)$ exactly or certify that $V(H) = \emptyset$.

The solver may also return a certified UNSAT core $U \subseteq H$. Any later hitting set containing U is UNSAT and does not require another solver call.

Algorithm 3 Factor branching

Require: A set of polynomials $P = \{p_1, \ldots, p_m\}$ and an exact branch solver $\mathcal{B}$.

$C \leftarrow \emptyset$ ▷ Certified UNSAT cores
$S \leftarrow \emptyset$ ▷ Solution-set representations
$R \leftarrow \textsc{Refine}(P, \emptyset)$
for all $H \in R$ **do**
    if H contains an UNSAT core from C **then**
        continue
    end if
    Use $\mathcal{B}$ to solve H
    if $\mathcal{B}$ returns a representation of $V(H)$ **then**
        Add the returned representation to S
    else
        if $\mathcal{B}$ returns a certified UNSAT core **then**
            Add the core to C
        end if
    end if
end for
return S, C

The order of the terminal hitting sets affects running time but not correctness. Optionally, we can sort the hitting sets by their estimated cost to solve. We use the number of generators in a hitting set as a proxy for the cost. This ordering can help us find many UNSAT cores quickly, which may be used to prune more difficult hitting sets.

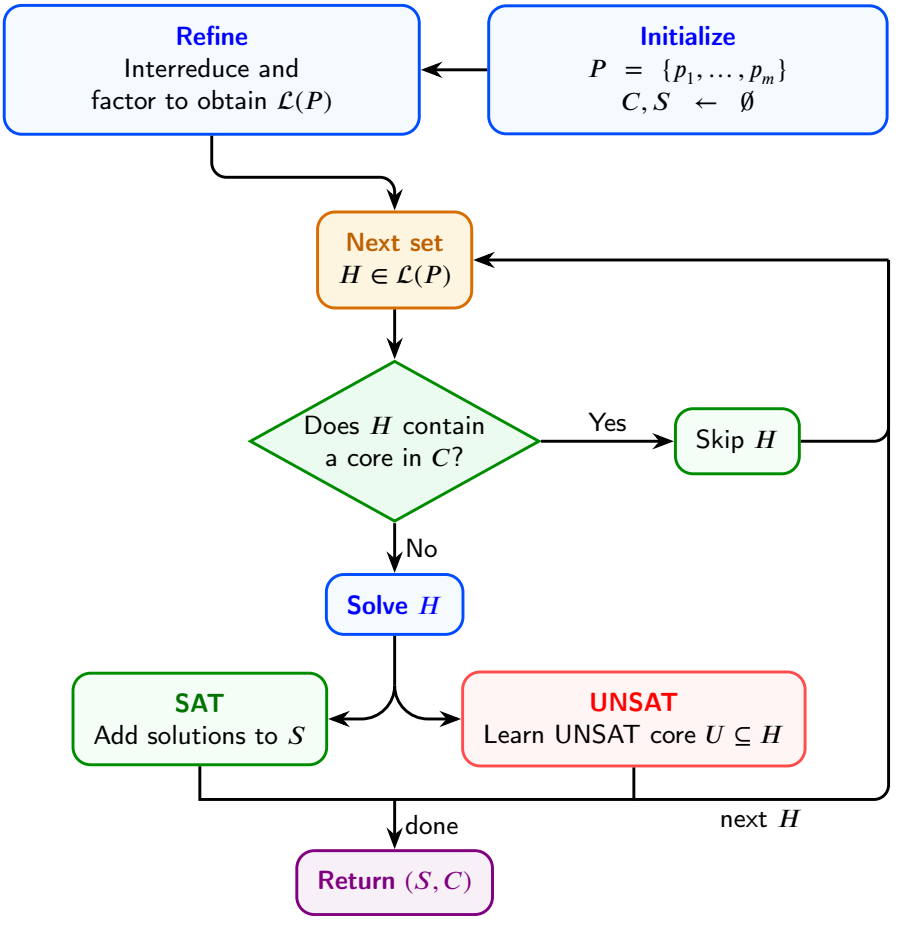


Figure 1: Factor branching procedure of Algorithm 3. Refinement produces terminal hitting-set branches, which the exact solver either represents or certifies as UNSAT.

3.2. Theoretical analysis

Let S be a collection of solution-set representations, and $\mathrm{Sol}(S)$ be the union of the represented sets. The theoretical analysis does not require K to be algebraically closed. If a system has no solution over the algebraic closure, it has no solution over K.

For a system Q, write $\mathcal{H}(Q)$ for the minimal hitting sets obtained in one factorization round.

Theorem 1 (One-round branch decomposition).

$$V(Q) = \bigcup_{H \in \mathcal{H}(Q)} V(H).$$

Proof. Write $Q = \{q_1, \ldots, q_m\}$ and let F_i be the nonconstant irreducible factors of q_i. Take $x \in V(Q)$. Since $q_i(x) = 0$, at least one factor in F_i vanishes at x for every i. The set of all factors that vanish at x is a hitting set of $F_1, \ldots, F_m$. Because the factor sets are finite, this set contains a minimal hitting set H. Every factor in H vanishes at x, so $x \in V(H)$.

For the reverse inclusion, take $x \in V(H)$ for some $H \in \mathcal{H}(Q)$. The set H intersects every F_i, so each q_i has a factor that vanishes at x. Hence $q_i(x) = 0$ for every i, and $x \in V(Q)$. □

Lemma 1 (Recursive refinement). *Suppose refinement of Q terminates with the finite collection $\mathcal{L}(Q)$. Then*

$$V(Q) = \bigcup_{H \in \mathcal{L}(Q)} V(H).$$

PROOF. Consider the finite recursion tree produced by Algorithm 2. For every internal node G, let $\widetilde{G}$ be its interreduction (or G itself when interreduction is disabled). Interreduction preserves the generated ideal, so $V(G) = V(\widetilde{G})$. Theorem 1 then gives

$$V(G) = \bigcup_{H \in \mathcal{H}(\widetilde{G})} V(H),$$

where the sets in $\mathcal{H}(\widetilde{G})$ are precisely the children of G. A bottom-up induction on the finite tree shows that the variety of each node equals the union of the varieties of the terminal nodes in its subtree. Applying this statement at the root Q proves the result. □

Theorem 2 (Termination of refinement). *Algorithm 2 terminates on every finite polynomial system.*

PROOF. If interreduction is disabled, every child consists of irreducible polynomials. Factoring such a child again returns it unchanged, so the condition $H = \widetilde{Q}$ makes it terminal.

Suppose interreduction is enabled. For an interreduced system Q, let

$$M(Q) = \langle \mathrm{LM}(q) : q \in Q \rangle.$$

For every hitting-set child H and every $q \in Q$, some $h \in H$ divides q. Hence $\mathrm{LM}(h)$ divides $\mathrm{LM}(q)$ and $M(Q) \subseteq M(H)$.

Suppose equality holds. For any $h \in H$, choose $q \in Q$ having h as a factor. Equality gives some $q' \in Q$ such that $\mathrm{LM}(q') \mid \mathrm{LM}(h) \mid \mathrm{LM}(q)$. Because Q is interreduced, this is possible only for $q' = q$. The two outer leading monomials are therefore equal, so the remaining factor of q is a unit. Thus h and q are associates, and their monic normalization gives $h = q$. Applying this argument to the factor selected for every $q \in Q$ gives $H = Q$, which is terminal.

It remains to account for the next interreduction. In an elementary reduction, the old leading monomial either remains, or it is already divisible by the leading monomial of the reducer. Consequently, interreduction cannot shrink the ideal generated by the current leading monomials. Thus, for every nonterminal child H and its interreduction $\widetilde{H}$,

$$M(Q) \subsetneq M(H) \subseteq M(\widetilde{H}).$$

Therefore M strictly increases along every recursive path.

The ascending-chain condition for monomial ideals rules out an infinite path. Every node has finitely many children, so König's lemma implies that the refinement tree is finite. Therefore Algorithm 2 terminates. □

The definition of an UNSAT core immediately justifies branch pruning.

Lemma 2. *Let U be an UNSAT core. If $U \subseteq H$, then $V(H) = \emptyset$.*

PROOF. Since $U \subseteq H$, every solution of H would be a solution of U. Hence $V(H) \subseteq V(U) = \emptyset$. □

The following theorem allows the use of Algorithm 3 without loss of correctness.

Theorem 3. *Algorithm 3 is sound and complete for P.*

PROOF. By Theorem 2, refinement terminates with a finite collection $\mathcal{L}(P)$. Every interreduction round preserves the current ideal, and every factorization round preserves the zero set by Theorem 1. Lemma 1 therefore gives

$$V(P) = \bigcup_{H \in \mathcal{L}(P)} V(H).$$

Consider any $H \in \mathcal{L}(P)$. If the algorithm prunes H, then it contains a certified UNSAT core, so $V(H) = \emptyset$ by Lemma 2. Otherwise, the exact solver either returns an exact representation of $V(H)$, which the algorithm adds to S,

or certifies that $V(H) = \emptyset$. Empty branches therefore make no contribution, while every represented branch contributes exactly its variety. Hence

$$\mathrm{Sol}(S) = \bigcup_{H \in \mathcal{L}(P)} V(H) = V(P).$$

Every representation added to S is exact, which proves soundness, and the equality proves completeness. Every member of C is certified by construction. □

Finally, we show that the factor branching algorithm terminates.

Theorem 4. *If the exact solver is known to terminate, then Algorithm 3 terminates.*

PROOF. By Theorem 2, the call $R = \textsc{Refine}(P, \emptyset)$ terminates and returns a finite collection. Therefore Algorithm 3 calls the exact solver a finite number of times, and each call terminates by assumption. Thus Algorithm 3 terminates. □

4. Recursive splitting

4.1. Factor branching with variable zeroing

Suppose the solver does not terminate within a reasonable amount of time on a hitting set produced by Algorithm 3.

Then we can further simplify the system by recursive splitting. In each child branch, we assume a side condition and run Algorithm 3 on the simplified system. This process continues until the solver terminates or the computation budget is exhausted.

For completeness, the side conditions must cover all possibilities. They should also simplify the system for the solver input and work well with Algorithm 2.

In this work, we use *variable zeroing*: recursive splitting with the side conditions $x = 0$ and $x \neq 0$ for a selected variable x. Our experience with the physical systems considered here suggests that a variable is often always zero or always nonzero.

Algorithm 4 Factor branching with variable zeroing

Require: A polynomial system P and a timeout τ
function ZEROINGBRANCHING(P, τ)
  $X \leftarrow$ VARIABLES(P)
  return SOLVEBYZEROING($P, X, \emptyset, \emptyset, \tau, \emptyset$)
end function
function SOLVEBYZEROING(Q, X, Z, N, τ, C)
  $(S, T) \leftarrow$ FACTORBRANCHINGWITHTIMEOUT(Q, τ, C)
  $U \leftarrow \emptyset$ ▷ Unresolved branches
  for all $R \in T$ **do**
    if $X = \emptyset$ **then**
      $U \leftarrow U \cup \{(R, Z, N)\}$
    else
      $x \leftarrow$ CHOOSEVARIABLE(R, X)
      $R_0 \leftarrow$ SUBSTITUTE($R, x, 0$)
      $R_1 \leftarrow$ UPDATESATURATION($R, N \cup \{x\}$)
      $(S_0, U_0, C) \leftarrow$ SOLVEBYZEROING($R_0, X \setminus \{x\}, Z \cup \{x\}, N, \tau, C$)
      $(S_1, U_1, C) \leftarrow$ SOLVEBYZEROING($R_1, X \setminus \{x\}, Z, N \cup \{x\}, \tau, C$)
      $S \leftarrow S \cup S_0 \cup S_1$
      $U \leftarrow U \cup U_0 \cup U_1$
    end if
  end for
  return (S, U, C)
end function

Algorithm 4 uses a global set C of learned UNSAT cores.

Let N be the variables declared nonzero on the current path. Subsection 2.4 describes two ways to encode these conditions as polynomial equations.

Solutions of the nonzero branch are projected onto the original variables, so the auxiliary variable does not appear in the output. The zero and nonzero decisions remain part of the solution-set representation.

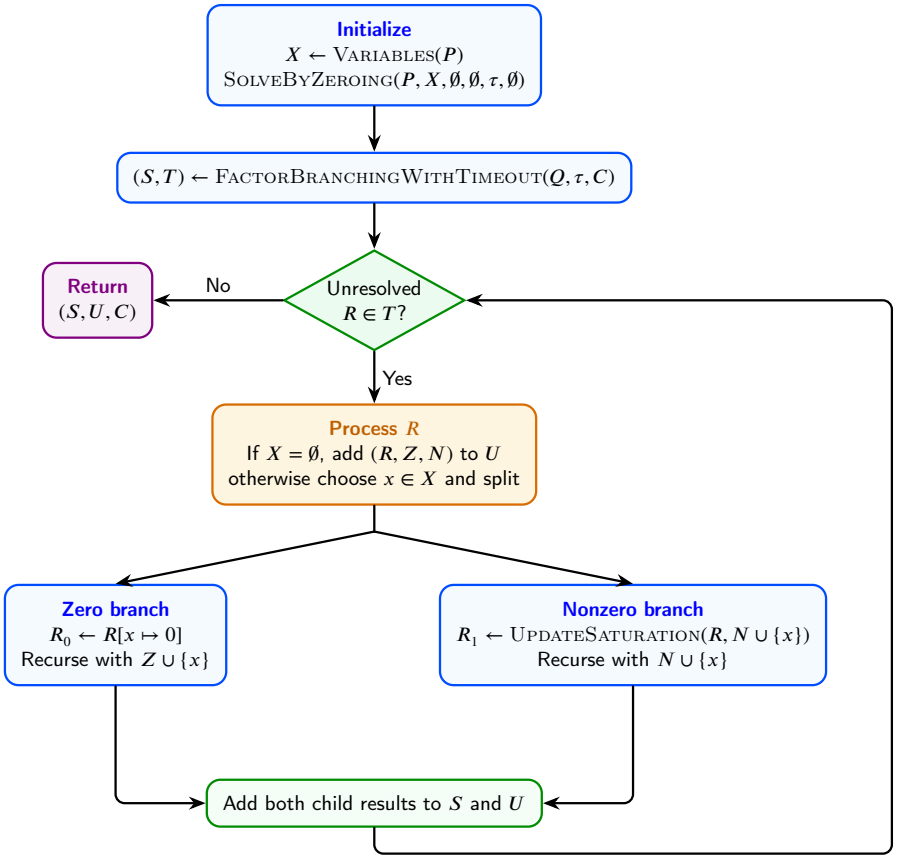


Figure 2: Flowchart of the recursive variable-zeroing procedure in Algorithm 4.

For comparison, we repeat the experiment twice: once with the fixed side conditions $x = 1$ and $x \neq 1$, and once with each split independently drawing an integer a uniformly from $\{-10, \ldots, 10\}$ and using the side conditions $x = a$ and $x \neq a$. Both runs have a ten-minute wall-clock limit. Neither alternative finishes within ten minutes, and in both cases the search has branches that remain unresolved even after exhausting every variable.

These comparisons demonstrate that zero/nonzero splits are substantially more effective than arbitrary affine splittings on these structured systems.

4.2. Choice of variable for splitting

The choice of variable for splitting affects the efficiency of the algorithm. If a variable barely occurs in the current hitting set, recursive calls may repeat the same work. A variable that occurs in many generators is a more promising choice. Setting it to zero removes all terms containing it at once. In the nonzero branch, it is appended to the product in Equation (2).

For each candidate variable x that is still available on the current branch, define

$$g_Q(x) = \#\{q \in Q : x \text{ occurs in } q\},$$
$$t_Q(x) = \#\{(q, m) : q \in Q,\ m \text{ is a term of } q,\ \deg_x(m) > 0\},$$
$$e_Q(x) = \sum_{q \in Q} \sum_{\substack{m \text{ a term of } q, \\ \deg_x(m) > 0}} \deg_x(m).$$

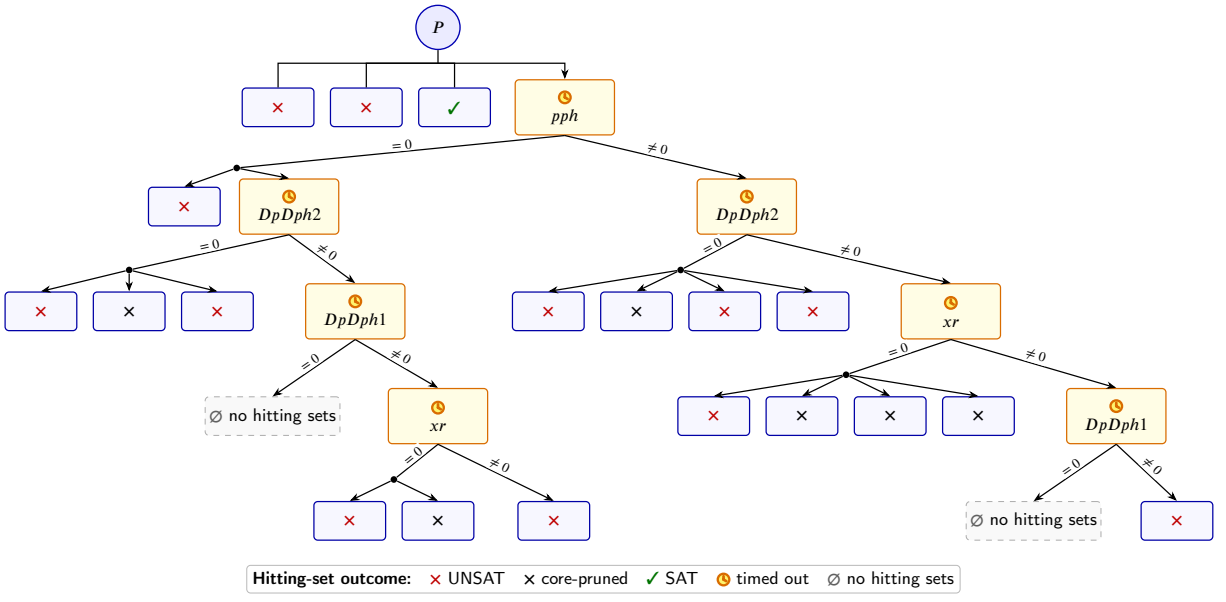


Figure 3: Recursion tree of factor branching with variable zeroing on the fifth-order graviton QFT system (detailed in Section 5.3) using M4GB with a five-second timeout per hitting set. The run takes 71 seconds.

Its score is the triple

$$h_Q(x) = \big(g_Q(x), t_Q(x), e_Q(x)\big),$$

and scores are compared lexicographically.

4.3. Branch contraction

Let $\mathcal{H}_0$ and $\mathcal{H}_1$ be the minimal hitting sets for $x = 0$ and $x \neq 0$. Suppose, up to exchanging the two children, that

$$\mathcal{H}_0 = \emptyset \qquad \text{and} \qquad \mathcal{H}_1 = \{H\}.$$

The first child is UNSAT. Theorem 1 and the variable-zeroing split give

$$V_Y(Q; Z, N) = V_Y(H; Z, N \cup \{x\}).$$

There is no need to run the branch solver on H before choosing the next splitting variable.

4.4. Theoretical analysis

The correctness argument uses only that the side conditions cover all possibilities. Let Y be the original variables, let A be the conditions on the current branch, and let $V_Y(Q; A)$ be the projection onto Y of the solutions of Q satisfying A. For variable zeroing, we write $V_Y(Q; Z, N)$ for the conditions $z = 0$ for $z \in Z$ and $y \neq 0$ for $y \in N$.

Theorem 5 (Correctness of a covering split). *Let $D_1, \dots, D_k$ be side conditions such that every point satisfying A satisfies at least one D_i. If Q_i encodes Q together with D_i, then*

$$V_Y(Q; A) = \bigcup_{i=1}^{k} V_Y(Q_i; A \cup \{D_i\}).$$

PROOF. Every solution on the left satisfies some D_i and therefore belongs to the corresponding set on the right. Every set on the right imposes all equations and conditions from the parent branch, so the reverse inclusion also holds. □

For variable zeroing, the conditions $x = 0$ and $x \neq 0$ cover all values of x. With $Q_0 = Q|_{x=0}$ and $Q_1 = \text{UpdateSaturation}(Q, N \cup \{x\})$, the theorem gives

$$V_Y(Q; Z, N) = V_Y(Q_0; Z \cup \{x\}, N) \cup V_Y(Q_1; Z, N \cup \{x\}). \tag{3}$$

If factor branching with a timeout returns solved representations S and timed-out hitting sets T, the proof of Theorem 3 gives

$$V_Y(Q; A) = \text{Sol}(S) \cup \bigcup_{R \in T} V_Y(R; A). \tag{4}$$

Theorem 6 (Correctness of recursive splitting). *Suppose every split uses side conditions that cover all possibilities. If the recursive procedure returns (S, U, C) on input P, where U contains the unresolved pairs (R, A_R), then*

$$V(P) = \text{Sol}(S) \cup \bigcup_{(R, A_R) \in U} V_Y(R; A_R).$$

PROOF. Equation (4) accounts for the solved and timed-out hitting sets at each call. An unresolved hitting set is either added to U or replaced by child branches. Theorem 5 shows that any covering family of child conditions preserves its solution set. Applying this argument recursively proves the result. □

Corollary 1 (Completeness of variable zeroing). *If $U = \emptyset$ in the output of Algorithm 4, then* $\text{Sol}(S) = V(P)$.

Theorem 7 (Termination of variable zeroing). *Algorithm 4 terminates if each timed solver call returns after τ.*

PROOF. Algorithm 2 terminates by Theorem 2 and returns finitely many hitting sets. Each variable-zeroing split removes one variable from the finite set X, so every recursive path is finite. Therefore the whole recursion tree is finite. □

5. Experiments

5.1. Setup

We implemented Algorithm 3 in Rust 1.97.1 and used Symbolica 1.5 for polynomial arithmetic over the rationals (Ruijl, 2023). In all experiments, we solve the branches with a Gröbner-basis computation. The available implementations are M4GB, F5, GVW, F4, and Singular 4.4.1 (Decker et al., 2025; Faugère, 1999, 2002; Gao et al., 2016; Makarim and Stevens, 2017). A basis containing 1 certifies that the branch is UNSAT. We obtain an UNSAT core by iterative refinement as in Section 5 of (Sun et al., 2016). The comparison also includes six direct whole-ideal Gröbner-basis configurations, three decomposition methods from Singular, Macaulay2 1.26.06, and a numerical method from HomotopyContinuation.jl under Julia 1.12.6 (Bezanson et al., 2017; Breiding and Timme, 2018; Decker et al., 2025; Grayson and Stillman, 2026).

For Algorithm 4, we test M4GB, F5, GVW, F4, and Singular as branch solvers, disable UNSAT-core learning, and impose a three-second limit on each hitting-set computation. We selected this cutoff by trying one, two, three, and five seconds on the hard gated systems. Three seconds is the smallest tested value that lets Singular complete the Katsura-8 residual (about 2.6 seconds); five seconds completes no additional hard system and only delays the subsequent splits.

We ran the benchmark on an Apple M4 Pro with 14 CPU cores and 24 GB of memory. Every table entry is a single run. Every method received 60 seconds per system; the three-second variable zeroing limit is an inner, per-hitting-set limit within that budget.

For factor branching and variable zeroing, T is end-to-end wall time and F is the time to first SAT branch, measured from the start of the minimal-hitting-set search. Factorization and input processing are therefore included in T but not in F. The complete measurements are stored in the appendix.

5.2. Baseline systems

The baseline has 20 generated systems in six families.

- Signed-distance cycles and affine-factor hypergraphs test many inexpensive hitting sets.

- The overlapping random multiples use seeded dense quadratic factors that recur across equations, with no factor shared by every equation.
- For gated systems, we create a new variable and multiply every equation in Cyclic, Katsura, or dense quadratic by it (Björck and Fröberg, 1991; Böge et al., 1986).
- This produces one easy selector hitting set and one nonlinear residual hitting set.

Family	Tested sizes	Variables/equations; hitting sets
Signed-distance cycle	$n = 8, 10, 12$	$8/8, 10/10, 12/12$; $8, 8, 32$
Overlapping random multiples	$(n, p, k) = (4, 6, 2), (6, 8, 3), (8, 10, 3)$	n/n; $4, 16, 28$
Positive affine hypergraph	$(n, m, k) = (8, 6, 4), (10, 8, 5), (12, 10, 6)$	n/m; $63, 239, 833$
Gated Cyclic	orders $5, 6, 7$	$6/5, 7/6, 8/7$; 2 each
Gated Katsura and dense quadratics	Katsura $6, 7, 8$; dense $n = 5, 6, 7$	$8/7, 9/8, 10/9, 6/5, 7/6, 8/7$; 2 each
Square affine hypergraph	$(n, k) = (8, 4), (10, 5)$	n/n; $31, 304$

Table 1
The generated systems used for the baseline benchmark.

The overlapping-random-multiple runs disable interreduction so that the generated factor-incidence pattern reaches the hitting-set enumerator unchanged. All other factor branching runs use interreduction. Table 2 selects the fastest completed factor branching backend and the fastest completed method in each baseline class. A factor branching entry gives T/F in seconds; the other entries give total wall time. "Timeout" means that every tested method in that class timed out.

System	Best factor branching T/F	Best direct GB	Best decomposition	Numerical
Signed distance, $n = 8$	F4 0.008/0.000	F4 0.008	facstd 0.190	32.410
Signed distance, $n = 10$	F4 0.025/0.002	F4 0.029	facstd 0.430	29.944
Signed distance, $n = 12$	F4 0.087/0.002	F4 0.472	primdec 3.056	36.880
Overlapping $(4, 6, 2)$	F5 0.008/0.000	F4 0.413	facstd 0.265	30.069
Overlapping $(6, 8, 3)$	F5 0.132/0.001	timeout	timeout	timeout
Overlapping $(8, 10, 3)$	Singular 2.931/0.010	timeout	timeout	timeout
Positive affine $(8, 6, 4)$	F5 0.008/0.000	timeout	facstd 0.368	50.934
Positive affine $(10, 8, 5)$	F5 0.020/0.000	timeout	facstd 1.836	timeout
Positive affine $(12, 10, 6)$	F5 0.067/0.000	timeout	facstd 53.318	timeout
Gated Cyclic-5	F4 0.009/0.000	F4 0.008	facstd 0.181	35.054
Gated Cyclic-6	Singular 0.045/0.008	Singular 0.309	M2 2.620	37.314
Gated Cyclic-7	timeout	timeout	timeout	58.287
Gated Katsura-6	Singular 0.049/0.008	GVW 0.226	facstd 0.233	31.532
Gated Katsura-7	Singular 0.283/0.008	GVW 3.176	facstd 0.539	31.980
Gated Katsura-8	Singular 2.657/0.052	timeout	facstd 3.596	34.264
Gated dense, $n = 5$	Singular 0.029/0.008	M4GB 0.171	facstd 0.198	29.289
Gated dense, $n = 6$	Singular 0.546/0.008	M4GB 4.736	facstd 0.755	30.053
Gated dense, $n = 7$	Singular 42.020/0.017	timeout	facstd 48.820	32.261
Square affine $(8, 4)$	Singular 0.470/0.023	timeout	facstd 0.601	timeout
Square affine $(10, 5)$	F4 1.174/0.000	timeout	facstd 2.141	timeout
#systems solved	19	10	17	14
#fastest (ties counted)	17	2	0	2

Table 2
Best completed baseline results across solver classes. All times are in seconds.

Factor branching is a good choice when factorization produces branch ideals that are substantially easier than the expanded input ideal. A large number of hitting sets can still be manageable when branch solving and core checks are cheap; the positive affine family shows that the hitting-set count alone is not decisive. The method is less effective when one hitting set retains the difficulty of the original nonlinear system, as Cyclic-7 does.

The measurements of time to first SAT branch support this explanation. On positive affine $(12, 10, 6)$, F rounds to 0.000 seconds despite the 833 hitting sets. On gated Katsura-8, the first SAT branch is found in 0.052 seconds, but the full run takes 2.657 seconds. Factorization therefore finds the easy component quickly, while the nonlinear residual controls completion time.

Variable zeroing is intended for a narrower case: an individual hitting set has timed out, but splitting a variable into zero and nonzero cases simplifies the system in both cases. The baseline suite gives limited evidence for that mechanism because none of the completed runs ever triggered a split ($O/S = 0/0$ in all cases; complete baseline zeroing results are provided in Table 5 in the Appendix). On these systems, variable zeroing behaves identically to factor branching with negligible framework overhead. On Cyclic-7, variable zeroing continues the stalled search and returns a partial result, but it does not complete the residual branch.

5.3. Physical Systems

The system is an order-five approximation constraining graviton couplings in asymptotic-safety models to remain consistent with quantum field theory (QFT) (Rafie-Zinedine, 2018; Vacca and Zanusso, 2010). Solving this system helps determine admissible fixed-point structures in quantum gravity (Vacca and Zanusso, 2010).

For this larger system, variable zeroing uses a one-second limit per hitting set and a 60-second total runtime limit. Factor branching uses a 60-second limit per hitting set and a 65-second external hard stop. Thus, an earlier completed hitting set can make the reported factor branching wall time slightly exceed 60 seconds.

Backend	Factor branching T/F (s)	Factor branching result	Variable zeroing T/F (s)	Variable zeroing result
M4GB	60.793/0.003	partial: 32 SAT, 1 timeout; $d = 14$	*timeout*/1.052	partial: 7 comp., 52 splits, 11 unresolved
F5	61.422/0.003	partial: 32 SAT, 1 timeout; $d = 14$	*timeout*/2.059	partial: 13 comp., 63 splits, 9 unresolved
GVW	61.653/0.003	partial: 32 SAT, 1 timeout; $d = 14$	*timeout*/2.071	partial: 13 comp., 60 splits, 9 unresolved
F4	60.218/0.003	partial: 32 SAT, 1 timeout; $d = 14$	*timeout*/1.051	partial: 7 comp., 56 splits, 12 unresolved
Singular	60.342/0.039	partial: 32 SAT, 1 timeout; $d = 14$	*timeout*/0.566	partial: 1 comp., 44 splits, 12 unresolved
`facstd`	1.067/0.013	complete: 32 SAT, 1 UNSAT; $d = 14$	0.824/0.077	complete: 6 comp., 0 splits, 0 unresolved

Table 3
Physical-system results from the fresh single-run benchmark.

When applied directly to the un-decomposed physical system, all tested standalone baselines timed out. Within our framework, the `facstd` backend is the only tested backend that completes its run on the physical system, doing so with both factor branching and variable zeroing in around one second. The five Gröbner-basis backends recover all SAT branches found by the complete `facstd` run. They time out on the remaining branch, which `facstd` certifies as UNSAT. Thus, they recover the full solution set but do not independently certify its completeness.

6. Conclusion

We introduced a solver-agnostic framework that decomposes polynomial systems into smaller branches by enumerating minimal hitting sets of their factor sets. The framework reuses certified UNSAT cores to prune branches and can recursively split timed-out branches into the cases $x = 0$ and $x \neq 0$.

We proved that refinement terminates and preserves the full solution set. Variable zeroing also preserves the solution set and terminates provided that each timed solver call returns. The output is complete whenever no branches remain unresolved.

On the baseline benchmarks, factor branching completed 19 of 20 systems, compared with 10 for direct Gröbner-basis methods, 17 for decomposition methods, and 14 for the numerical method. On the QFT system, `facstd` was the only tested backend to certify a complete result; the other backends recovered the SAT branches but did not certify completeness.

Current experiments provide limited evidence about when variable zeroing improves completion. Future work will develop more effective splitting strategies and more efficient ways to generate and reuse UNSAT cores.

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A. Complete experimental results

The tables below give every run summarized in Table 2. Unless stated otherwise, entries are end-to-end wall times in seconds. P marks a partial run and TO an outer timeout. These are single measurements.

A.1. Factor branching

Table 4 reports all five branch-solver configurations. A partial entry found the easy component shown in the main text but left at least one hitting set unresolved.

A.2. Variable zeroing

Table 5 gives the complete output of the fastest completed variable zeroing configuration evaluated across the baseline suite. Every backend receives three seconds per hitting set and a 60-second outer budget, and UNSAT-core learning is disabled. The overlapping-random-multiple rows disable interreduction, as in the factor branching

System	M4GB	F5	GVW	F4	Singular
Signed distance, $n = 8$	0.651	0.053	0.122	0.008	0.083
Signed distance, $n = 10$	0.046	10.927	45.797	0.025	0.084
Signed distance, $n = 12$	0.182	6.256	1.841	0.087	0.271
Overlapping $(4, 6, 2)$	0.017	0.008	0.011	0.015	0.038
Overlapping $(6, 8, 3)$	1.053	0.132	0.280	0.657	0.201
Overlapping $(8, 10, 3)$	TO	10.937	30.687	TO	2.931
Positive affine $(8, 6, 4)$	0.009	0.008	0.010	0.008	0.497
Positive affine $(10, 8, 5)$	0.021	0.020	0.028	0.020	1.870
Positive affine $(12, 10, 6)$	0.072	0.067	0.096	0.068	6.498
Gated Cyclic-5	0.010	0.010	0.011	0.009	0.022
Gated Cyclic-6	P 60.010	5.960	2.275	P 60.013	0.045
Gated Cyclic-7	P 60.875	P 60.504	P 60.955	P 60.024	P 60.103
Gated Katsura-6	0.366	0.483	1.338	2.540	0.049
Gated Katsura-7	4.487	15.220	42.113	P 60.012	0.283
Gated Katsura-8	54.055	P 60.325	P 60.047	P 60.012	2.657
Gated dense, $n = 5$	0.177	0.071	0.200	0.204	0.029
Gated dense, $n = 6$	4.716	3.194	7.475	21.498	0.546
Gated dense, $n = 7$	P 60.008	P 60.050	P 60.041	P 60.015	42.020
Square affine $(8, 4)$	15.117	TO	TO	3.096	0.470
Square affine $(10, 5)$	14.750	TO	47.561	1.174	2.548

Table 4
End-to-end wall times for every factor-branching configuration.

experiment. Here N is the number of recursion nodes, E the number of hitting-set evaluations, O/S the numbers of per-hitting-set timeouts and variable zeroing splits, and C/U the numbers of returned components and unresolved branches. The time entry is T/F.

The dense $n = 7$ run failed while constructing the first recursive child: the expanded Singular program exceeded the operating system's command-line argument limit. This is an implementation error, not an UNSAT result or a timeout. We also tried M4GB, F5, GVW, and F4 on this system. These backends execute in-process; after a per-hitting-set timeout their non-cancellable worker threads continued alongside the recursive children until the process was killed under memory pressure. Singular remains preferable for the hard rows because its external worker can be terminated. The Cyclic-7 run exhausted all useful splitting variables on one route and therefore returned one unresolved branch.

A.3. Direct whole-ideal Gröbner bases

A.4. Decomposition and numerical methods

System	Backend	T/F	N	E	O/S	C/U	Result
Signed distance, $n = 8$	F4	0.008/0.001	1	8	0/0	8/0	complete
Signed distance, $n = 10$	F4	0.025/0.002	1	8	0/0	8/0	complete
Signed distance, $n = 12$	F4	0.091/0.003	1	32	0/0	32/0	complete
Overlapping $(4, 6, 2)$	F5	0.009/0.001	1	4	0/0	4/0	complete
Overlapping $(6, 8, 3)$	F5	0.132/0.021	1	16	0/0	16/0	complete
Overlapping $(8, 10, 3)$	Singular	2.892/0.152	1	28	0/0	28/0	complete
Positive affine $(8, 6, 4)$	Singular	0.181/0.012	1	4	0/0	4/0	complete
Positive affine $(10, 8, 5)$	F4	0.060/0.007	1	242	0/0	242/0	complete
Positive affine $(12, 10, 6)$	Singular	9.600/0.078	1	511	0/0	511/0	complete
Gated Cyclic-5	F4	0.011/0.001	1	2	0/0	2/0	complete
Gated Cyclic-6	Singular	0.043/0.009	1	2	0/0	2/0	complete
Gated Cyclic-7	Singular	24.118/0.009	15	17	8/7	1/1	partial
Gated Katsura-6	Singular	0.049/0.009	1	2	0/0	2/0	complete
Gated Katsura-7	Singular	0.284/0.009	1	2	0/0	2/0	complete
Gated Katsura-8	Singular	2.588/0.009	1	2	0/0	2/0	complete
Gated dense, $n = 5$	Singular	0.029/0.008	1	2	0/0	2/0	complete
Gated dense, $n = 6$	Singular	0.535/0.008	1	2	0/0	2/0	complete
Gated dense, $n = 7$	Singular	4.797/–	–	–	–	–	error
Square affine $(8, 4)$	Singular	0.445/0.012	1	31	0/0	31/0	complete
Square affine $(10, 5)$	F4	1.326/0.015	1	304	0/0	304/0	complete

Table 5
Selected variable-zeroing result for every system. Times are end-to-end wall time and time to first SAT branch in seconds.

System	M4GB	F5	GVW	F4	Singular	Macaulay2
Signed distance, $n = 8$	0.646	0.014	0.010	0.008	0.333	0.570
Signed distance, $n = 10$	0.130	0.221	0.112	0.029	0.194	0.518
Signed distance, $n = 12$	3.266	20.044	8.685	0.472	0.512	1.293
Overlapping $(4, 6, 2)$	1.000	24.865	0.472	0.413	TO	TO
Overlapping $(6, 8, 3)$	TO	TO	TO	TO	TO	TO
Overlapping $(8, 10, 3)$	TO	TO	TO	TO	TO	TO
Positive affine $(8, 6, 4)$	TO	TO	TO	TO	TO	TO
Positive affine $(10, 8, 5)$	TO	TO	TO	TO	TO	TO
Positive affine $(12, 10, 6)$	TO	TO	TO	TO	TO	TO
Gated Cyclic-5	0.012	0.279	0.009	0.008	0.183	0.480
Gated Cyclic-6	TO	TO	1.738	TO	0.309	0.917
Gated Cyclic-7	TO	TO	TO	TO	TO	TO
Gated Katsura-6	0.436	44.961	0.226	5.277	14.430	TO
Gated Katsura-7	5.165	TO	3.176	TO	TO	TO
Gated Katsura-8	TO	TO	TO	TO	TO	TO
Gated dense, $n = 5$	0.171	8.884	0.179	0.200	0.529	TO
Gated dense, $n = 6$	4.736	TO	6.755	17.472	TO	TO
Gated dense, $n = 7$	TO	TO	TO	TO	TO	TO
Square affine $(8, 4)$	TO	TO	TO	TO	TO	TO
Square affine $(10, 5)$	TO	TO	TO	TO	TO	TO

Table 6
End-to-end wall times for every direct whole-ideal Gröbner-basis configuration.

System	facstd	primdecGTZ	Macaulay2	Homotopy
Signed distance, $n = 8$	0.190	0.320	1.020	32.410
Signed distance, $n = 10$	0.430	0.667	3.224	29.944
Signed distance, $n = 12$	4.800	3.056	25.223	36.880
Overlapping $(4, 6, 2)$	0.265	TO	TO	30.069
Overlapping $(6, 8, 3)$	TO	TO	TO	TO
Overlapping $(8, 10, 3)$	TO	TO	TO	TO
Positive affine $(8, 6, 4)$	0.368	TO	TO	50.934
Positive affine $(10, 8, 5)$	1.836	TO	TO	TO
Positive affine $(12, 10, 6)$	53.318	TO	TO	TO
Gated Cyclic-5	0.181	0.337	0.709	35.054
Gated Cyclic-6	2.729	TO	2.620	37.314
Gated Cyclic-7	TO	TO	TO	58.287
Gated Katsura-6	0.233	TO	TO	31.532
Gated Katsura-7	0.539	TO	TO	31.980
Gated Katsura-8	3.596	TO	TO	34.264
Gated dense, $n = 5$	0.198	TO	TO	29.289
Gated dense, $n = 6$	0.755	TO	TO	30.053
Gated dense, $n = 7$	48.820	TO	TO	32.261
Square affine $(8, 4)$	0.601	TO	TO	TO
Square affine $(10, 5)$	2.141	TO	TO	TO

Table 7
End-to-end wall times for every decomposition and numerical configuration.